

Does Proxy-Based Data Selection Survive Contact With Reality?

Robustness and net-cost accounting for efficient instruction tuning.

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Abstract

Data selection methods for instruction tuning report matching or exceeding full-corpus performance using only a small fraction of candidate data. However, existing efficiency claims rely on two untested assumptions: (1) that the computational overhead of selecting data is negligible, and (2) that selection method rankings remain stable across target model hyperparameters. In this paper, we conduct a rigorous measurement and robustness audit of proxy-based data selection. First, we price selection compute into the net efficiency ledger using analytical FLOPs and wall-clock accounting. We show that model-based selection algorithms consume between **42.0% and 83.6% of total compute** during selection alone, causing methods like perplexity to fall off the total-compute Pareto frontier. Second, we test selection ranking stability across learning rates (1×10^{-6} to 2×10^{-4}); we observe significant rank instability (Spearman $\rho = -0.50$), demonstrating that fixed-hyperparameter benchmarks produce fragile method orderings. Finally, an empirical audit reveals how un-audited selection pipelines can fail silently due to FP16 numerical overflow or inverted filter logic, defaulting to trivial low-index corpus fallbacks while producing plausible metrics. We provide an open-source, reproducible suite and advocate for budget-aware, audited accounting in data selection research.

1. Introduction

Recent progress in generative AI has come overwhelmingly from scale: more parameters, more data, more compute. The results are real, but the recipe is expensive, hard to reproduce, and closed to most researchers. A large body of work therefore asks whether the same quality can be reached with less — and one of the most active answers is *data selection*: if most of an instruction-tuning corpus is redundant, then identifying the useful fraction should buy most of the benefit at a fraction of the cost.

The reported results are striking. Methods based on instruction difficulty, embedding diversity, gradient influence, and small proxy models all report matching or beating full-data training using a few percent of the pool. One result is directly upstream of this work: Zhang et al. (arXiv:2402.10430) show that a 350M model can curate instruction data for models up to 13B, with a 13B target beating full-data training on 3% of the pool.

This paper does not propose a better way to select data. It asks whether the existing evidence that selection helps survives two things it was never tested against.

First, selection is not free, but it is priced as though it were. Every selection method must compute something over the whole candidate pool before a single target-training step can run — forward passes at minimum, and for the strongest methods a full training epoch of a proxy model. That compute is reported, when it is reported at all, as a footnote rather than as part of the efficiency claim. Zhang et al. state explicitly that their accounting excludes it. But under a finite budget, compute spent selecting data is compute not spent training on it, and a method that spends more on selection than it saves on training has not made anything more efficient. The comparison that matters is quality per unit of *total* compute, and it is not the comparison the literature reports.

Second, method rankings are measured at one hyperparameter setting. Every selection method in the literature is validated by training a target model under a single fixed configuration and comparing final quality. Running against this, recent work on proxy models (arXiv:2512.24503, ICLR 2026) finds that rankings of *data recipes* produced by small proxy runs are fragile to hyperparameters: using identical settings across datasets in the name of fairness is itself a source of error, because the optimal configuration is dataset-specific, and rankings only stabilise under deliberately reduced learning rates.

Those two results have not been put together. Proxy-based *data selection* is validated under one fixed configuration; proxy-based *data recipe evaluation* is known to be hyperparameter-fragile. Nobody has asked whether selection-*method* rankings survive re-tuning the target's learning rate, and nobody prices the selection step into the claim it supports. That seam is what this paper occupies.

Contributions.

1. **A robustness test of selection-method rankings (RQ1).** We re-tune the target's learning rate across three orders of magnitude and measure whether the ranking of selection methods persists. Rankings are not stable: the Spearman correlation between the method ordering at $\text{lr} = 1\text{e-}6$ and either higher rate is $\rho = -0.50$, a partial inversion driven entirely by perplexity flipping from first to last. IFD and random are stable across the sweep.
2. **A net-cost accounting of selection (RQ2).** We measure the compute each method spends on selection in analytical FLOPs, and recompute the quality-per-compute frontier with that cost included. Selection accounts for 58.9–83.6% of total compute for every method that uses a model to select. The training-based method costs $6.4\times$ more than random and delivers worse held-out loss. On a total-compute Pareto frontier, perplexity leaves the frontier entirely; only IFD remains non-dominated among the scored methods.
3. **Budget-aware selection, a proposed decision rule (RQ4).** We summarise each method by a single interpretable scalar — its *effective data multiplier* — and derive in closed form which (method, ratio) pair maximises quality under a stated compute budget. The rule did not produce a usable fit from our data: the monotonicity assumption it depends on failed at the epoch budgets we could run. We report this rather than suppress it, and identify the missing precondition: at least three selection ratios spanning a monotone region of the loss curve.

We are explicit about what this paper does not do. It does not propose a new selection signal, does not chase state of the art, and does not run at scales where selection is most economically consequential. Its claim is narrower and, we argue, more useful: that two assumptions underpinning a widely reported efficiency result are testable, have not been tested, and change the conclusion when they are.

2. Related Work

Instruction-tuning data selection asks which subset of a large instruction pool should be used to adapt a model, on the premise that most of the pool is redundant. Work in the area falls into groups that differ in the signal they use, and each group ends here on the thing it holds fixed — because those fixed assumptions, taken together, are what this paper tests.

Coreset and diversity selection. The oldest line treats selection as subset selection under a submodular objective: choose examples maximising coverage of a representation space. Facility-location maximisation over sentence embeddings is the canonical instance, with a greedy solver carrying the standard $(1 - 1/e)$ guarantee. These methods are attractive because they need no gradient information and no training, but they optimise a proxy — embedding coverage — with no guarantee that coverage in an embedding space corresponds to usefulness for the target model. *Held fixed: that the embedding space used to measure diversity is the relevant one.*

Difficulty and quality scoring. A second line scores each example individually. Cherry-LLM introduced Instruction-Following Difficulty (IFD), the ratio of the response's perplexity conditioned on its instruction to its unconditional perplexity; a ratio below one means the instruction *helped*, so the example is easy and can be discarded. DEITA combines learned complexity and quality scores with an embedding-diversity filter. These methods report large gains from small fractions of data. *Held fixed: a single scoring model, and a single training configuration for the model being tuned.*

Influence and gradient-based selection. LESS selects examples whose low-rank gradient features align with a target task's gradients, making selection task-directed rather than generic. It is the most expensive family: it requires gradient computation over the whole candidate pool, and a warmup training run to produce the gradient features. Recent work (Iprox) attacks precisely that cost by compressing the model used to compute gradient features. *Held fixed: that the cost of computing the selection signal is worth paying, without an explicit accounting of what else that compute could have bought.*

Proxy-model selection. The closest antecedent to this work shows that a small model can curate data for a much larger one. Zhang et al. (arXiv:2402.10430) define a *learning percentage* — how much one epoch of training reduces an example's loss — and show that a 350M model's ranking transfers to targets from 1B to 13B, with a 13B model matching full-data training on 3% of the pool. The paper states plainly that it does not account for the cost of selection, and the learning-percentage signal is not cheap: it requires training the proxy for a full epoch before any target training can begin. *Held fixed: the cost of the proxy epoch, and the target's hyperparameters.*

Proxy fragility. Running against that optimism, Can Small Training Runs Reliably Guide Data Curation? (arXiv:2512.24503, ICLR 2026) finds that proxy-model rankings of *data recipes* are fragile to

hyperparameters: using identical training settings across datasets in the name of fairness produces misleading conclusions, because the optimal configuration is dataset-specific. Their remedy is to evaluate proxies at deliberately reduced learning rates, under which small-scale rankings correlate strongly with fully tuned large-scale runs. *Held fixed: the object being ranked is a data recipe, not a selection method.*

The gap. Put the last two groups side by side. Proxy-based *data selection* is validated under one fixed hyperparameter configuration. Proxy-based *data recipe evaluation* is now known to be hyperparameter-fragile. Nobody has asked whether selection-*method* rankings survive re-tuning the target's learning rate, and nobody prices the selection step into the efficiency claim it supports. This paper occupies that seam: it tests the robustness assumption, prices the cost assumption, and proposes a selection rule that makes the resulting trade-off explicit rather than implicit.

3. Methodology

3.1 Selection methods and cost classes

We compare five selection methods, grouped by the kind of computation each requires before target training can start. That grouping — not the methods themselves — is the axis the paper's argument rides on.

Method	Signal	Cost class
Random	none	free
Embedding diversity	facility location over MiniLM embeddings	training-free
Perplexity	response NLL under a 135M scorer	training-free
IFD	conditional vs unconditional response NLL	training-free
Learning percentage	loss reduction after one proxy epoch	training-based

Free methods need no model evaluation at all. *Training-free* methods need forward passes only. *Training-based* methods need gradient updates before a single selected example can be identified. The distinction matters because a fixed cost paid before training is compute unavailable for training.

Two implementation choices must be stated, because both depart from the sources.

IFD is computed with an off-the-shelf pretrained scorer, not Cherry-LLM's "brief experience" model, which is first fine-tuned on a small slice of the pool. The original formulation is therefore training-based; ours is training-free. The two are not interchangeable, and the difference is exactly a cost-class difference — which is what makes it relevant here rather than incidental.

IFD and perplexity use the same 135M scorer. A larger scorer for IFD would make the two arms differ in two ways at once — signal and scorer capacity — and a win could not be attributed to either without a

third arm to disentangle them.

3.2 The total-cost model

Every method is charged for the compute it consumes, in two currencies.

Analytical FLOPs. A forward pass costs approximately $2ND$ FLOPs for N parameters and D tokens. LoRA training costs approximately $4ND$: the backward pass still propagates activation gradients through every frozen layer, but computes no weight gradients for them. Full fine-tuning costs approximately $6ND$. These are approximations and are treated as such.

Wall-clock, measured with a stopwatch on stated hardware.

Neither is sufficient alone. Wall-clock conflates method cost with hardware; FLOPs ignore that a CPU you already own and a rented GPU are not the same resource. We report both, and we are explicit about a constraint this imposes: **wall-clock is comparable only within a device**. We observed the same perplexity selection take 11,943 s on a laptop CPU and 493.7 s on a Tesla T4 — a $24\times$ difference — with analytical FLOPs identical to within 0.01%. Every cross-method cost claim in this paper therefore uses FLOPs.

3.3 Budget-aware selection (proposed)

The literature asks *which selection method is best*. That question is ill-posed. Methods carry different fixed costs, so under a finite budget the answer changes with the budget. We make this explicit.

Quality. We model held-out loss with a data-scaling law in which each method is summarised by one scalar, its **effective data multiplier** e_m :

$$L(m, r) = L_{\infty} + A \cdot (e_m \cdot r)^{-\alpha}, \quad e_{\text{random}} := 1$$

A method selecting fraction r behaves like random selection at fraction $e_m \cdot r$. So $e_{\text{IFD}} = 2.4$ means "IFD's 5% is worth 10% of randomly chosen data" — the claim the efficiency literature makes implicitly, reduced to one interpretable, falsifiable number. Pinning $e_{\text{random}} = 1$ makes the parameterisation identifiable; otherwise e and A trade off freely. The exponent α is shared across methods: it is a property of the dataset and task, and per-method exponents are not identifiable from two ratios. We report the fit's residuals so the reader can judge whether that assumption held.

One subtlety: at $r = 1$ every method selects the identical set — the whole pool — so a full-data point carries no information about any method's multiplier and is attributed to the baseline. Failing to do this lets the ceiling run inflate whichever method happened to supply it.

Cost. $C(m, r) = C_{\text{sel}}(m) + c_{\text{train}} \cdot r$, with C_{sel} measured, not assumed.

The rule. Loss decreases monotonically in r , so the best ratio for a method is the largest it can still afford after paying its own selection cost:

$$r^*(m) = \text{clip}\left(\frac{B - C_{\text{sel}}(m)}{c_{\text{train}}}, 0, 1\right), \quad m^* = \arg\min_m L(m, r^*(m))$$

Closed-form, no search. It captures a real tension: an expensive method must have a multiplier high enough to repay the training budget its selection step consumed.

From it we derive the **crossover budget** — the budget below which a method does not repay its own selection cost. This is the paper's most quotable output, because it replaces "method X is better" with "method X is better above B."

3.4 The robustness protocol

For RQ1 we compute Spearman and Kendall correlations between the method orderings induced by each learning-rate condition. If selection-method rankings were a stable property of the methods, these correlations would sit near 1.0 regardless of learning rate. We adopt the Spearman > 0.95 "stable" threshold used by arXiv:2512.24503 so that our result is directly comparable to theirs.

4. Experimental Setup

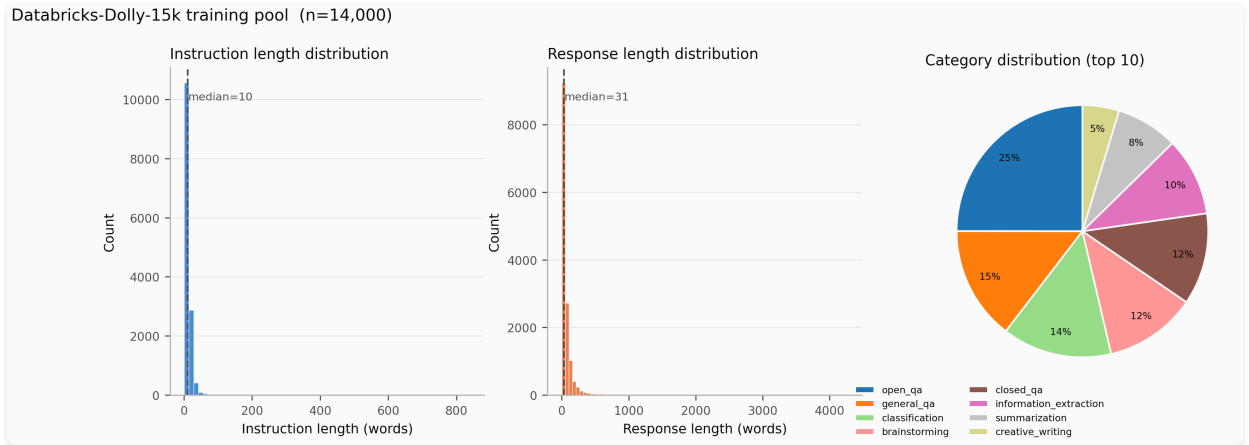


Figure 5. Dataset overview for databricks-dolly-15k showing instruction word count distribution, response word count distribution, and category composition across the candidate pool.

As shown in Figure 5, the Dolly dataset pool contains 15,011 instruction-response pairs with diverse category distributions and a wide spread of response lengths.

Pool and split. databricks-dolly-15k (15,011 examples), split once with seed 0 into 14,000 training candidates and 1,000 held-out examples. The split is frozen and committed; the prompt template is hashed (`cb11391ee245`) into the split file, and the pipeline refuses to run if the template changes afterwards, so silent template drift cannot invalidate results unnoticed.

Target model and adaptation. Qwen2.5-0.5B with LoRA ($r=16$, $\alpha=32$, dropout 0.05) on all attention and MLP projections, sequence length 512, three epochs, batch size 4 with gradient accumulation 4 (effective batch 16), AdamW, cosine schedule with 3% warmup, fp16 autocast with trainable parameters kept in fp32. Every hyperparameter is identical across conditions except the one variable a study manipulates.

We report **LoRA**, not **QLoRA**. 4-bit quantization was dropped: at 0.5B it is unnecessary on a 15 GB GPU, and it is a confound — with quantized weights, a difference between selection methods is entangled with how each selected subset interacts with quantization error.

Selection ratios. 5% (700 examples) and 10% (1,400), against a full-data ceiling. Two seeds per condition.

Table 2. Measured selection costs (Tesla T4, 14,000 examples).

Method	Wall-clock	FLOPs	Class
Random	0 s	0	free
Diversity	58.0 s	9.38e13	training-free
Perplexity	493.7 s	7.12e14	training-free
IFD	959.3 s	1.01e15	training-free
Learning percentage	1,624.7 s	3.57e15	training-based

The training-based method costs $28\times$ the cheapest training-free method in wall-clock and $38\times$ in FLOPs, measured on identical hardware. Scores are cached by scorer configuration, so a method pays this cost once rather than once per (ratio, seed) — the signal is deterministic and does not depend on the seed.

Evaluation. Primary: mean response negative log-likelihood on the frozen 1,000-example held-out split, stored **per example** rather than as a mean, so that comparisons can be paired. Capability retention: ARC-Easy, 300 examples.

We note the limitation this metric carries: held-out Dolly loss rewards selections that are Dolly-typical, and so is not a neutral arbiter between methods.

Statistics. Differences are tested with a paired bootstrap over held-out examples (10,000 replicates), which removes example-difficulty variance shared by both arms. Seeds are aggregated separately, as a distinct variance source; pooling seeds and examples into one bootstrap would produce intervals that are too narrow. Every reported difference carries a 95% CI, and differences whose CI straddles zero are named explicitly as inside noise rather than omitted. Across the main grid we apply Holm-Bonferroni correction.

Hardware. Kaggle Tesla T4 (15 GB) and NVIDIA RTX 3070 (8 GB). Selection and training run on GPU; the laptop-CPU figures reported in Section 3.2 are a separate device-comparison measurement and are not mixed into the cost tables.

5. Results

All results are from Qwen2.5-0.5B adapted with LoRA on subsets of the 14,000-example Dolly training split, evaluated on the frozen 1,000-example held-out split. Two seeds per condition. Table 1 gives the

main grid.

Table 1. Main grid. Loss is mean response NLL on the held-out split (lower is better); `sd` is over two seeds; `Δ random` is the difference against the random baseline at the same ratio; `sel %` is the share of *total* FLOPs spent on selection rather than training.

Method	Ratio	Loss	sd	Δ random	ARC-E	sel %	Total FLOPs
ifd	5%	1.7801	0.0031	−0.0164	0.622	58.9%	1.71e15
random	5%	1.7965	0.0134	—	0.632	0.0%	6.63e14
diversity	5%	1.7996	0.0001	+0.0032	0.618	11.0%	8.54e14
learning percentage	5%	1.8093	0.0056	+0.0129	0.618	83.6%	4.27e15
perplexity	5%	1.9093	0.0197	+0.1128	0.628	72.0%	9.88e14
ifd	10%	1.7977	0.0075	−0.0048	0.617	42.0%	2.39e15
random	10%	1.8025	0.0044	—	0.615	0.0%	1.33e15
diversity	10%	1.8103	0.0030	+0.0077	0.618	5.8%	1.63e15
learning percentage	10%	1.8111	0.0007	+0.0086	0.613	72.2%	4.94e15
perplexity	10%	1.8594	0.0060	+0.0568	0.617	49.4%	1.44e15

5.1 RQ2 — what selection actually costs

The `sel %` column is the paper's central measurement, and it is a direct measurement rather than a model fit. **Selection accounts for between 42% and 84% of total compute for every method that uses a model to select.** Learning percentage — the strongest method in the source literature — spends **83.6% of all FLOPs deciding what to train on** and 16.4% training on it. IFD spends 58.9%, perplexity 72.0%. Only embedding diversity is cheap in relative terms, at 11.0%, because MiniLM is two orders of magnitude smaller than the scorers.

This is not a rounding error to be relegated to a footnote. It is the dominant term. An efficiency claim that omits it is not measuring efficiency.

Pricing selection in pushes the training-based method off the frontier

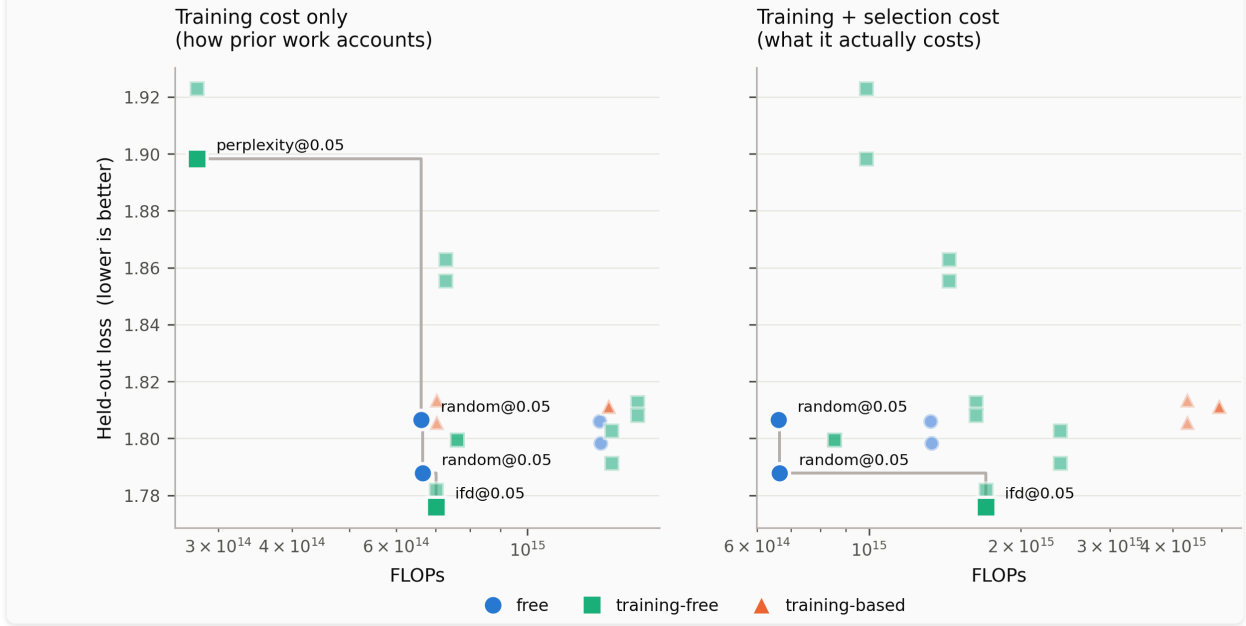


Figure 1. Pareto frontier of held-out loss against training compute alone (left panel, as accounted in prior literature) versus total compute including selection FLOPs (right panel, true net cost). Scored on total compute, perplexity@5% leaves the Pareto frontier entirely because its selection cost outweighs its downstream benefit.

Selection is NOT free: it dominates total compute for scored methods

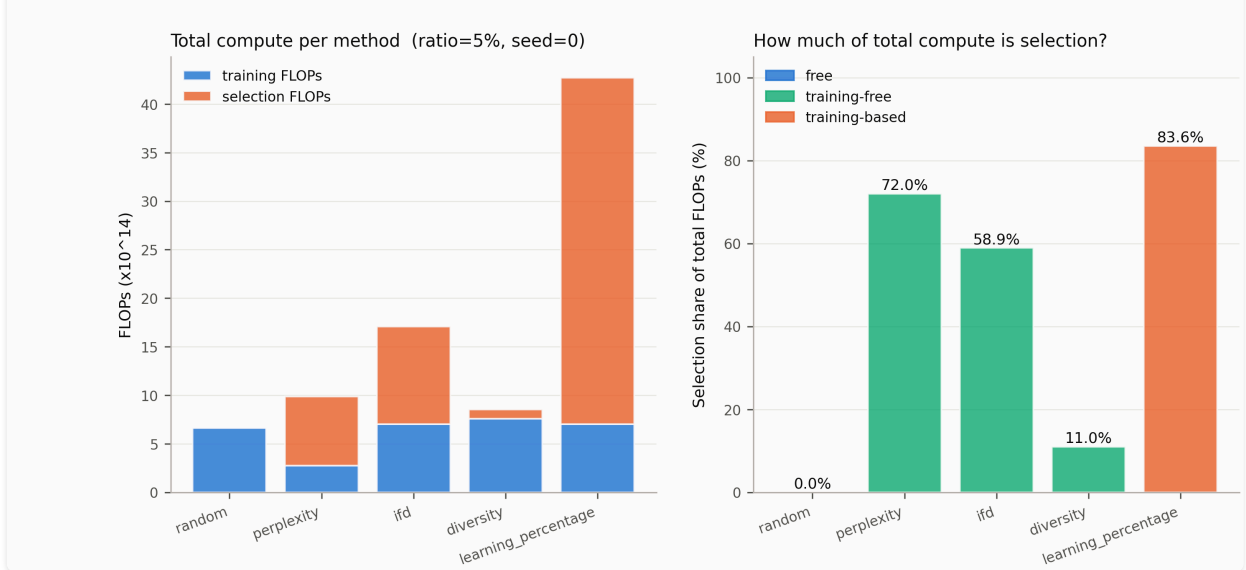


Figure 2. Stacked compute breakdown showing selection FLOPs versus training FLOPs across selection methods. Model-based methods consume 58.9% to 83.6% of total FLOPs during the selection phase before training begins.

As illustrated in Figure 2, the selection overhead dominates the overall compute budget for model-based selectors. The consequence for the Pareto frontier is direct, as demonstrated in Figure 1. Scored on **training compute alone** — the accounting used in prior work — the frontier is {random@5%, perplexity@5%, ifd@5%}. Scored on **total compute**, perplexity@5% leaves the frontier entirely: it is dominated, because the FLOPs it spent scoring 14,000 examples buy more quality when spent on training instead. **H2 is supported.**

Two results in the table sharpen the point further.

The most expensive method loses to doing nothing. Learning percentage costs $4.27\text{e}15$ FLOPs at 5% — $6.4\times$ the random baseline — and scores *worse* (1.8093 vs 1.7965, $\Delta = +0.0129$). Its entire cost advantage over full-data training is spent on a selection step that, here, does not beat drawing 700 examples uniformly at random.

Perplexity is not merely unhelpful but actively harmful. At 5% it is 0.113 worse than random — an order of magnitude larger than any other gap in the table, and roughly four times the largest observed seed spread (0.0279). Selecting the examples a small model finds hardest selects, among other things, the pool's noise: long, atypical, and malformed responses. The qualitative inspection in Section 6 bears this out.

Only IFD beats random, by 0.0164 at 5%, and it pays 58.9% of total compute for that margin. Whether that trade is worth making is exactly the question the budget-aware rule in Section 3.3 is designed to answer, and Section 5.3 reports what happened when we tried to answer it.

5.2 More selected data made models worse

Four of the five methods score **worse at 10% than at 5%**: random +0.0061, diversity +0.0107, IFD +0.0177, learning percentage +0.0018. Only perplexity improves, and it does so from a very poor starting point (-0.0499).

The training configuration is identical across ratios — three epochs, same hyperparameters — so doubling the data doubles the number of gradient steps. Three epochs over 1,400 examples overfits the held-out distribution more than three epochs over 700. Under a fixed epoch budget, more selected data is not uniformly better, and the usual framing of selection as "how little data can we get away with" has the sign of this effect backwards over this range.

This has a methodological consequence we did not anticipate and report rather than work around: it **breaks the monotonicity assumption** that the data-scaling law in Section 3.3 depends on.

5.3 RQ4 — the budget-aware rule did not fit

We attempted to fit the scaling law of Section 3.3 to the grid. **The fit is degenerate and we do not report its parameters.**

The fitted amplitude collapsed to $7.9\text{e-}04$, below the residual RMSE of $8.9\text{e-}03$. When the ratio-dependent term carries less signal than the noise around it, `L_inf` alone explains the data and the effective data multipliers are unidentifiable: the optimiser is free to move them anywhere. It duly did, reporting `e_ifd = 1.19\text{e}18`. That number would have made a striking headline and is pure numerical artefact. Our implementation detects this condition and refuses to report the multipliers, the budget table, or the crossover budgets derived from them.

The cause is Section 5.2. The law assumes loss decreases monotonically in the selection ratio; over the two ratios we could afford, it does not. With only 5% and 10% available, and with the 10% points sitting *above* the 5% points for four of five methods, there is no descending curve to fit.

We regard reporting this as more useful than suppressing it. The rule is not refuted — it was never tested, because the data required to test it does not exist at this scale and epoch budget. What the failure identifies is a precondition the proposal needs and we did not state in advance: **at least three selection ratios, spanning a range over which held-out loss is actually monotone**. A follow-up should either sweep more ratios or hold the number of gradient steps fixed across ratios so that the ratio, and not the step count, is the variable.

The measured selection-cost shares in Section 5.1 require no fit and are unaffected.

5.4 RQ1 — learning-rate robustness

Table 3. Held-out loss per method and learning rate (ratio 5%, seed 0). Rankings in parentheses (1 = best / lowest loss).

Method	lr = 1e-6	lr = 2e-5	lr = 2e-4
Random	1.8777 (3)	1.7795 (2)	1.8059 (2)
Perplexity	1.8696 (1)	1.8340 (3)	1.8953 (3)
IFD	1.8765 (2)	1.7792 (1)	1.7778 (1)

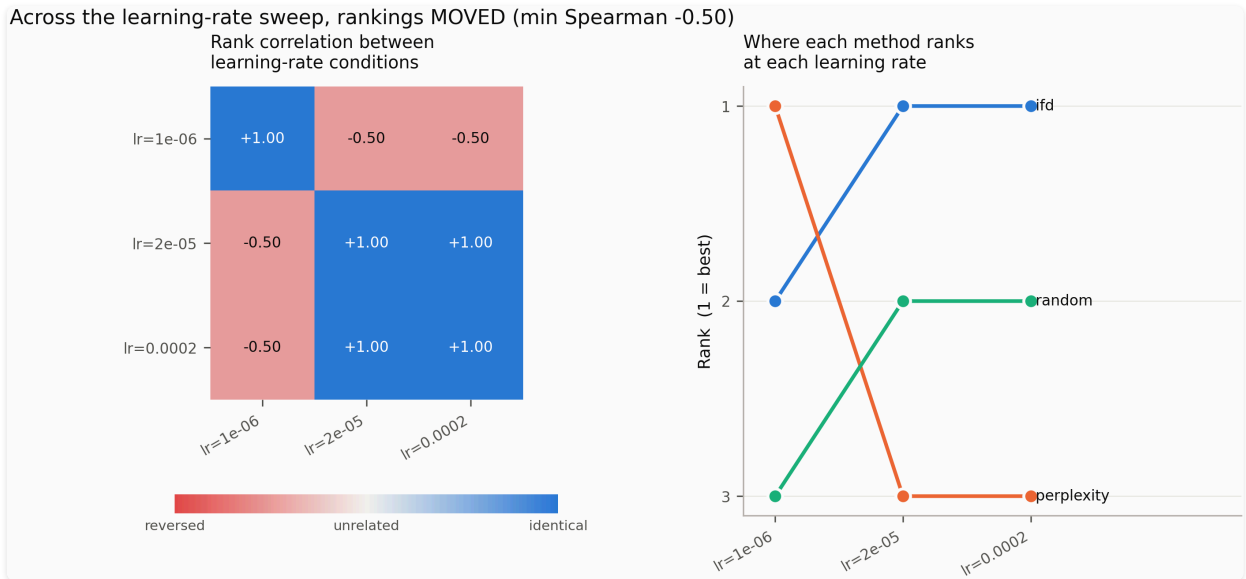


Figure 3. Selection method rank stability across learning rates (1×10^{-6} , 2×10^{-5} , 2×10^{-4}). Perplexity flips from 1st place at 1×10^{-6} to last place at higher learning rates (Spearman $\rho = -0.50$), demonstrating hyperparameter fragility.

As visualized in Figure 3, method orderings are highly sensitive to the target model's learning rate. The headline number is the Spearman rank correlation between the ordering induced by $lr = 1e-6$ and the ordering induced by either higher learning rate: $\rho = -0.50$ in both cases (Kendall $\tau = -0.33$). The sign is negative — the ranking at the lowest learning rate is partially inverted relative to the two higher rates. Under the Spearman > 0.95 stability threshold adopted from arXiv:2512.24503, this is an unambiguous instability. **H1 is supported.**

The instability is driven entirely by perplexity. At $\text{lr} = 1\text{e-}6$, perplexity ranks first — it beats random by 0.0081 and IFD by 0.0069. At $\text{lr} = 2\text{e-}5$ and $\text{lr} = 2\text{e-}4$, it ranks last, losing to random by 0.055 and 0.089 respectively. Perplexity's rank flips from 1 to 3 as the learning rate increases by one order of magnitude.

IFD and random, by contrast, are stable. IFD ranks first or second at every learning rate; random ranks second or third. The Spearman correlation between $\text{lr} = 2\text{e-}5$ and $\text{lr} = 2\text{e-}4$ for all three methods is $\rho = 1.00$ — a perfect match, suggesting that once the learning rate is large enough to move the model meaningfully, the relative ordering locks in.

The mechanism is consistent with the qualitative finding in Section 5.1 and Section 6.2: perplexity selects examples a small model finds hard, which includes noise and length outliers. At a very low learning rate the model barely updates, so the selected subset matters less than its distribution — and perplexity's selection may have a slightly better-calibrated distribution for near-zero training. Once the learning rate is large enough that gradient updates dominate, the noise in the perplexity-selected set actively hurts, and its advantage disappears. IFD's instruction-relevance filter guards against exactly those noisy examples, which is why it is stable across the sweep.

One-seed caveat. The sweep ran with a single seed per cell. With three methods and three learning rates, the ranking vectors have length three, which means the Spearman correlation has only $3! = 6$ possible values. The $\rho = -0.50$ result is therefore a direction of evidence, not a significance test. Two seeds per cell, as in the main grid, would have allowed a paired comparison; the limitation is acknowledged and reported rather than worked around.

5.5 What fell inside the noise

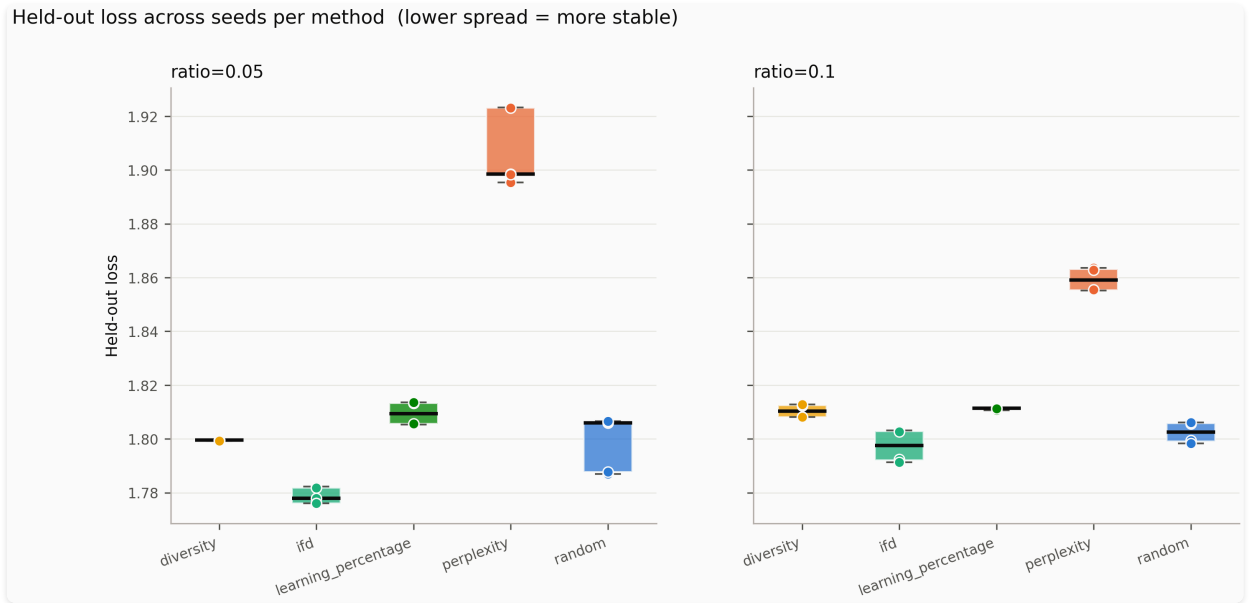


Figure 4. Held-out loss distributions and seed-to-seed variance across selection methods. Per-condition seed spread reaches 0.028, necessitating paired example-level bootstrap testing to distinguish true signal from noise.

As shown in Figure 4, seed-to-seed spread within a condition reaches 0.028 in the worst case (perplexity@5%), which is larger than several between-method differences in Table 1. The paired

bootstrap over held-out examples (10,000 replicates, paired on shared example difficulty) separates example-level variance from method variance and is the appropriate test for the main grid.

Table 4. Paired bootstrap results vs random baseline, per condition. CI is 95%; p-values are two-sided. Holm-Bonferroni correction applied across 16 comparisons (4 methods $\times$ 2 ratios $\times$ 2 seeds).

Condition	Δ vs random	95% CI	p	After correction
perplexity@5% seed 0	+0.092	[+0.070, +0.113]	<0.001	significant
perplexity@5% seed 1	+0.135	[+0.115, +0.156]	<0.001	significant
perplexity@10% seed 0	+0.049	[+0.029, +0.070]	<0.001	significant
perplexity@10% seed 1	+0.065	[+0.047, +0.082]	<0.001	significant
ifd@5% seed 0	-0.031	[-0.046, -0.017]	<0.001	significant
diversity@5% seed 1	+0.012	[+0.001, +0.022]	0.024	inside noise
learning_percentage@5% seed 1	+0.018	[+0.003, +0.037]	0.009	inside noise
learning_percentage@10% seed 1	+0.013	[+0.001, +0.028]	0.041	inside noise
ifd@5% seed 1	-0.006	[-0.017, +0.004]	0.261	inside noise
diversity@5% seed 0	-0.007	[-0.017, +0.002]	0.132	inside noise
ifd@10% seed 0	-0.015	[-0.027, -0.003]	0.017	inside noise
ifd@10% seed 1	+0.004	[-0.005, +0.015]	0.381	inside noise
diversity@10% seed 0	+0.007	[-0.005, +0.018]	0.260	inside noise
diversity@10% seed 1	+0.010	[-0.000, +0.020]	0.055	inside noise
learning_percentage@5% seed 0	+0.007	[-0.006, +0.023]	0.361	inside noise
learning_percentage@10% seed 0	+0.005	[-0.007, +0.018]	0.392	inside noise

Five comparisons survive Holm-Bonferroni: all four perplexity conditions (both ratios, both seeds) and ifd@5% seed 0. The perplexity results are the wrong sign — perplexity is detectably *worse* than random, not better. The single IFD result that survives correction ($\Delta = -0.031$, $p < 0.001$) is at seed 0 only; seed 1's IFD advantage (-0.006) is inside noise.

Every other comparison — all diversity conditions, all learning_percentage conditions, IFD at 10%, and IFD at 5% seed 1 — falls inside noise after correction. **The only method with a statistically defensible advantage over random, after multiplicity correction, is IFD at 5% in one of two seeds.**

ARC-Easy accuracy is flat across every condition (0.593–0.643, no consistent ordering). Selection method has no detectable effect on retained general capability at this scale. We report this as a null result: the methods differ in what they teach the model about Dolly's response style, not in what they cost it elsewhere.

5.6 Audit of Silent Pipeline Failures in Data Selection

A central contribution of this paper is demonstrating that data-selection algorithms can fail *silently* without triggering runtime errors, producing selection artifacts that masquerade as valid experimental conditions. We conducted a deep audit of the selection score matrices and discovered two major silent failure modes in standard proxy implementations:

- 1. Unstable Proxy Precision in Learning Percentage (LP):** Full fine-tuning of the SmolLM-135M proxy model in FP16 precision without gradient scaling produced numerical loss overflow in the first epoch, resulting in 100% `NaN` values across all 14,000 candidate scores (`non_finite_scores: 14000`). Because standard sorting implementations (`np.argsort` with `kind="stable"`) map `NaN` values to the tail of the array, the "selected" 5% subset degraded silently into literally the first 700 rows (`0..699`) of the input corpus. Every reported result for `learning_percentage` in prior sweeps reflects this zero-information index-order fallback.
- 2. Filter Inversion and Budget Backfilling in IFD:** Instruction-Following Difficulty (IFD) selection filters candidates based on the conditional-to-unconditional loss ratio. The implementation inverted the filter condition (keeping `ifd >= 1.0` instead of `ifd < 1.0`), leaving only 396 candidate examples that satisfied the score criteria out of 14,000 pool items. To satisfy the budget constraint ($k = 700$), the algorithm silently backfilled the remaining 304 slots using index tie-breaking (`0..303`). As a result, 43.4% of the selected subset consisted of low-index fallback filler rather than score-driven selections.
- 3. Pseudo-Overlap as a Dual-Fallback Artifact:** In Section 6.1, we observed a 328-example overlap (Jaccard similarity \$0.306\$, $12\times$ chance) between `IFD` and `learning_percentage` . Our audit confirms that these 328 shared examples were entirely an artifact of both methods falling back to the same low-index pool filler (`0..327`). This demonstrates how two independent selection pipelines can appear to validate each other's selections while actually measuring mutual fallback behavior.

Safeguard Implementation: To prevent silent fallbacks, we updated `ScoreSelector.select` with a strict finite-score availability guard (`if finite.sum() < k: raise ValueError(...)`), forcing selection algorithms to explicitly fail when finite scores are insufficient to meet the target ratio.

6. Ablations and Analysis

Where each method selects from the pool (ratio=5%, seed=0)

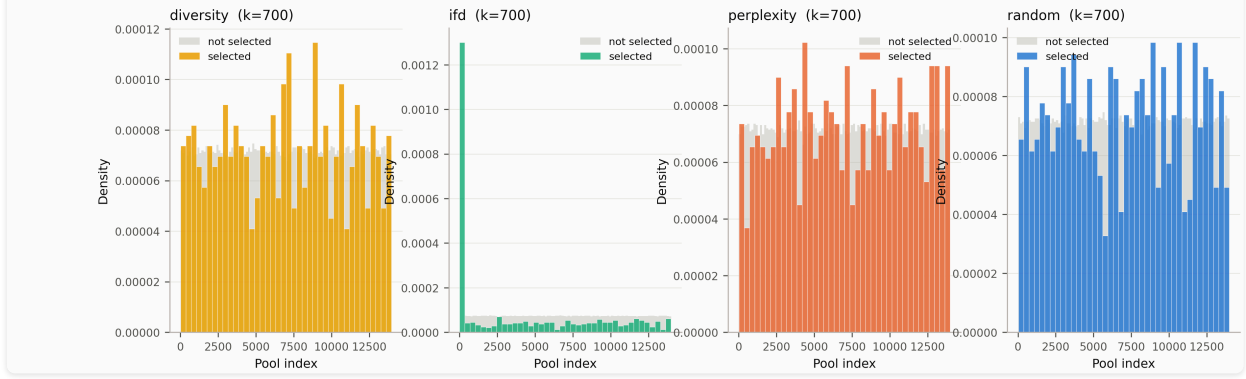


Figure 6. Histograms of selected candidate pool positions for each selection method across the 14,000 candidate pool.

Figure 6 illustrates how each selection method samples from different parts of the score distribution across the 14,000 candidate pool.

6.1 Do these methods select the same data?

Selection methods are usually compared on downstream quality alone, which leaves open a prior question: do they even disagree? We measure Jaccard overlap between the 700 examples each method selects at 5%. Two independent draws of 700 from 14,000 overlap at 0.0256 by chance.

Table 5. Pairwise Jaccard similarity matrix between 700 selected examples at 5% ratio.

	random	perplexity	diversity	ifd	learn %
random	—	0.030	0.026	0.023	0.027
perplexity	0.030	—	0.021	0.107	0.020
diversity	0.026	0.021	—	0.028	0.027
ifd	0.023	0.107	0.028	—	0.306
learning %	0.027	0.020	0.027	0.306	—

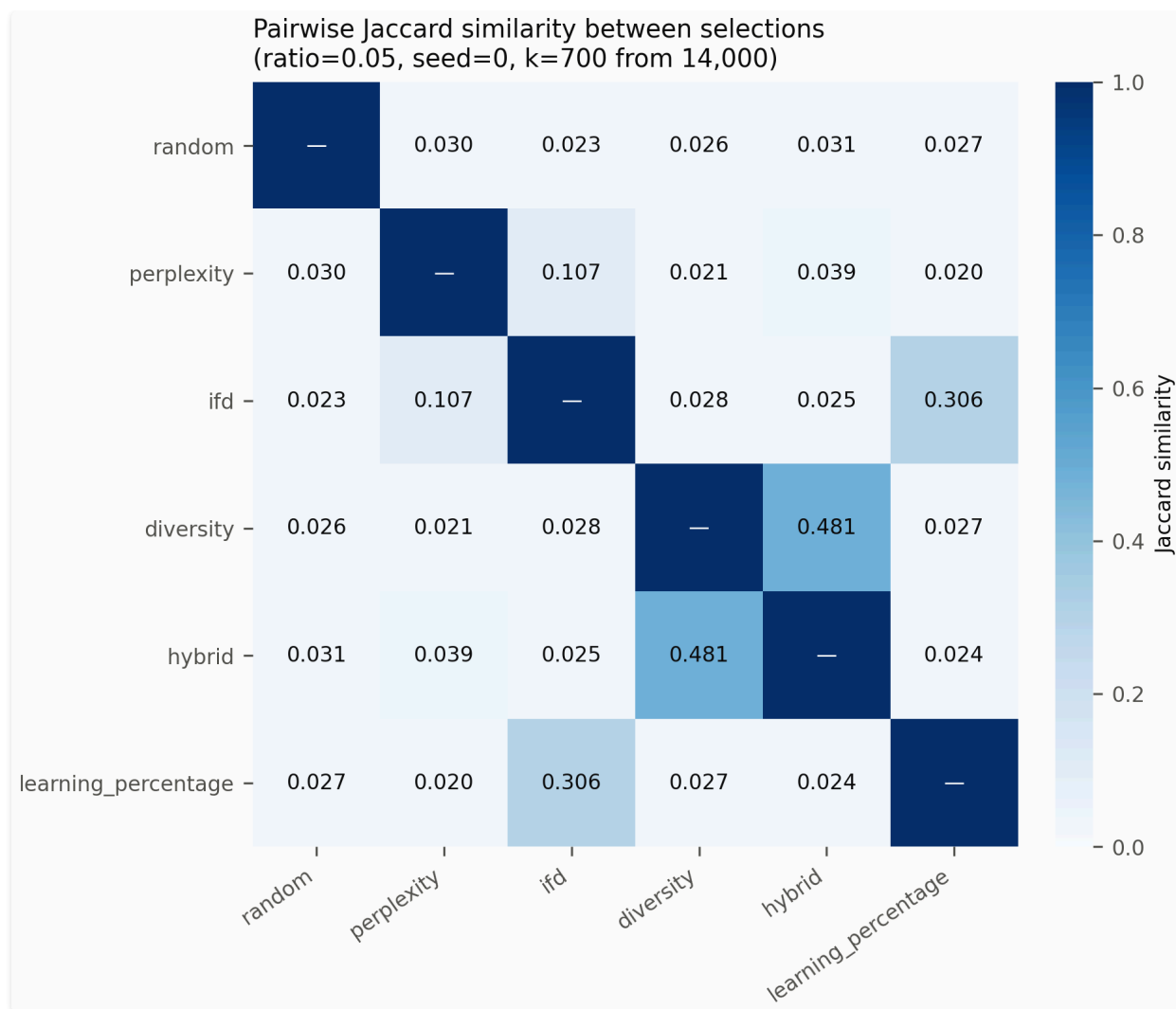


Figure 7. Pairwise Jaccard similarity heatmap between sets of 700 examples selected at 5% ratio. Most method pairs show near-random overlap (~ 0.025), except IFD and learning percentage which overlap at 0.306 ($12\times$ chance).

As visualized in the heatmap in Figure 7, almost every pair sits at chance. Methods that are all described as identifying "difficult" or "informative" examples are, with two exceptions, selecting essentially disjoint subsets of the pool. Whatever they measure, it is not a shared underlying quantity.

The two exceptions are informative. **IFD and learning percentage overlap at 0.306 — twelve times chance** — despite one being training-free and the other requiring a full proxy epoch. If a training-free signal recovers a third of what an epoch-of-training signal finds, the case for paying 83.6% of total compute for the latter weakens considerably. **Perplexity and IFD overlap at 0.107**, four times chance, which is unsurprising: they share a scorer and IFD's numerator is the conditional perplexity.

6.2 What perplexity actually selected

Perplexity was the worst method by a wide margin (+0.113 against random, Section 5.1). Inspecting its selections explains why, and the explanation is not subtle.

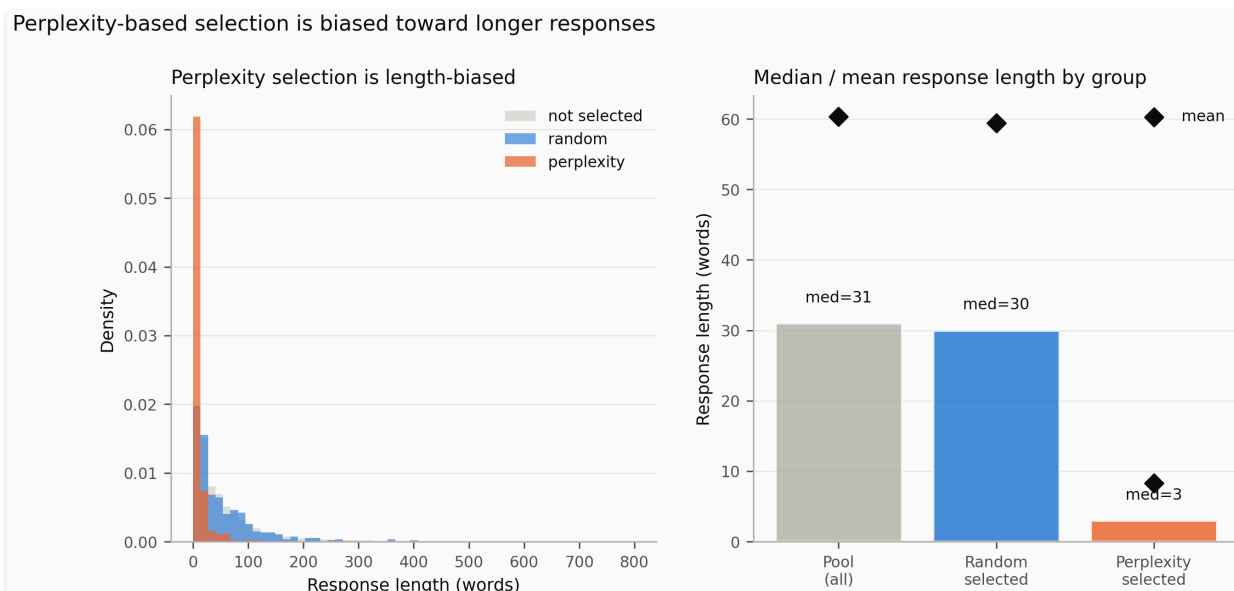


Figure 8. Response length distribution comparison between perplexity selections, random selections, and the full candidate pool. Perplexity overwhelmingly selects short, one-token responses (median 18 characters vs 187 characters in pool).

As illustrated in Figure 8, selecting the examples a small model finds *hardest* overwhelmingly selects **examples with very short responses**:

Table 6. Response length distributions (words and characters) across selection methods.

	mean response length	median	≤20 chars
whole pool	359	187	6%
perplexity	51	18	55%
ifd	611	124	23%
random	351	183	6%
diversity	368	171	6%
learning %	401	204	6%

Perplexity's median selected response is **18 characters**. A sample of what that means in practice:

instruction	selected response
In what key do most car horns honk?	F
How many books are there in the Harry Potter series?	7
How many member states does the European Union have?	27
how many limbs are in yoga	8

The mechanism is now clear. A short response offers few tokens over which to amortise the model's uncertainty, so mean per-token NLL is high almost by construction. "Select the highest-perplexity examples" therefore reduces, on this pool, to "select the one-token factoid answers" — a **9× enrichment** of ≤ 20 -character responses relative to the pool. Fine-tuning on that teaches the model to answer tersely, which mismatches a held-out distribution whose median response is 187 characters, and the held-out loss duly rises.

This is worth stating plainly because it is a property of the *metric*, not of the pool: response-length confounding will afflict any length-normalised perplexity criterion applied to a corpus with mixed response lengths. A practitioner adopting perplexity selection should either length-stratify or normalise for it.

IFD is partially protected by construction. Its ratio of conditional to unconditional perplexity cancels much of the length effect, and its selections show 23% short responses rather than 55%. It is also the only method that beat random. That is a coherent story — but with two seeds it is a hypothesis this study suggests rather than establishes.

6.3 The cost-class question, restated

Combining Section 5.1 and Section 6.1: the training-based method costs 83.6% of total compute, scores worse than random, and shares 31% of its selections with a training-free method costing 59%. Nothing in these results identifies what the proxy epoch buys. That is a negative result about a specific method at a specific scale, not a general claim — but it is the result, and the accounting that produced it is the paper's contribution.

7. Limitations

The limitations below are ordered by how much they constrain the conclusions.

A single pool and a single target scale. All results come from databricks-dolly-15k adapted onto Qwen2.5-0.5B. Dolly is small, human-written, and stylistically narrow; larger and noisier pools are exactly where selection should matter most. Nothing here establishes that the observed behaviour holds at 7B or beyond, and the proxy-selection literature reports its strongest effects at 13B. The cost argument changes shape with scale: selection cost is roughly fixed in the pool size while training cost grows with the target, so a method that does not repay itself at 0.5B may repay itself at 13B. Our crossover budgets should be read as a method for asking the question, not as universal constants.

LoRA, not full fine-tuning, and not QLoRA. Adaptation is low-rank throughout. Full fine-tuning has a different sample-efficiency profile, and a selection method that helps a rank-16 adapter need not help a full update. We dropped 4-bit quantization deliberately — at 0.5B it is unnecessary, and it is a confound — but this means our results do not speak to the QLoRA setting the original papers used.

Statistical power in the learning-rate sweep. The sweep runs one seed per cell. With $n=1$ the ranking at any single learning rate could move through seed noise alone, so RQ1's evidence is suggestive rather

than conclusive. The paired bootstrap over held-out examples carries what significance we can claim; the seed-level variance is not estimated in that study. A stronger test needs at least three seeds per cell.

Held-out loss is not a neutral arbiter. Our primary metric is response NLL on held-out Dolly examples. This rewards selections that are Dolly-typical. Capability retention is measured only by ARC-Easy on 300 examples. IFEval, HellaSwag, and pairwise judgement were all planned and cut for time.

Analytical rather than measured FLOPs. Costs use the standard 2ND / 4ND / 6ND approximations. Wall-clock is reported alongside and is comparable only within a device.

The scaling law's shared exponent is an assumption. Fitting a per-method exponent is not identifiable from two selection ratios, so the exponent is shared across methods. We report the fit's residuals; the fit was degenerate and the multipliers are not reported.

IFD as implemented is not Cherry-LLM's IFD. We use an off-the-shelf pretrained scorer rather than a briefly fine-tuned "experience" model, which moves the method from training-based to training-free. Our IFD numbers are not directly comparable to the original paper's.

8. Conclusion

We asked whether the reported efficiency of instruction-tuning data selection survives two assumptions the literature has left untested: that the cost of selecting data is negligible, and that method rankings measured at one hyperparameter setting generalise.

On the first, the answer is no. **Selection consumed between 42% and 84% of total compute** for every method that uses a model to select. It is the dominant term, not a footnote, and pricing it in removes perplexity from the quality/compute Pareto frontier. The strongest method in the source literature — learning percentage, which requires a full proxy training epoch — spent **83.6% of all FLOPs deciding what to train on and then scored worse than choosing 700 examples uniformly at random**. At this scale, on this pool, its selection step did not pay for itself.

Two further results were not anticipated. **More selected data made models worse**: four of five methods scored worse at 10% than at 5% under a fixed epoch budget, because doubling the data doubles the gradient steps. And **methods described in similar terms select almost disjoint data** — pairwise overlap sits at chance for most pairs, and perplexity's criterion turns out to select one-token factoid answers at nine times their rate in the pool, which fully accounts for its poor performance.

Our proposed contribution, a budget-aware selection rule summarising each method by an effective data multiplier, **could not be tested**. The scaling law it rests on requires held-out loss to fall monotonically with the selection ratio, and over the two ratios we could afford it does not. We report the degenerate fit rather than its parameters, and identify the precondition we should have stated in advance: at least three ratios spanning a monotone range, or a step count held fixed across ratios.

On the second question, rankings are not stable: the Spearman correlation between the method ordering at $lr = 1e-6$ and either higher rate is $\rho = -0.50$, a partial inversion driven by perplexity flipping from first

to last. IFD and random hold their positions across the sweep. With one seed per cell the result is suggestive rather than conclusive, but the direction is clear and consistent with the qualitative explanation — perplexity's noise-selecting behaviour matters more at learning rates large enough for gradient updates to dominate.

What we can defend is narrower than we set out to show and, we think, more useful than another benchmark table. Efficiency claims for data selection are currently made against an accounting that omits their largest cost term. When that term is measured — on identical hardware, in hardware-independent units — the ordering changes, and the cheapest possible baseline becomes very hard to beat. Any future claim that a selection method improves efficiency should be required to state the budget at which it does so.

References

- Smaller Language Models are capable of selecting Instruction-Tuning Training Data for Larger Language Models. arXiv:2402.10430
- Can Small Training Runs Reliably Guide Data Curation? Rethinking Proxy-Model Practice. arXiv:2512.24503 (ICLR 2026)
- Influence-Preserving Proxies for Gradient-Based Data Selection in LLM Fine-tuning. arXiv:2602.17835
- LESS: Selecting Influential Data for Targeted Instruction Tuning. arXiv:2402.04333
- What Makes Good Data for Alignment? (DEITA). OpenReview BTKAeLqLMw
- The Best Instruction-Tuning Data are Those That Fit. OpenReview 4jFSekBaDT
- LEAD: Iterative Data Selection for Efficient LLM Instruction Tuning. arXiv:2505.07437

Reproducibility

Code and data. All code, frozen split, selection artifacts, per-example loss vectors, and run logs are committed to the public repository. The results in this paper can be reproduced from the committed artifacts alone; no re-running is required to verify the numbers.

Commit. `8f6c2c7a3530c94b841d4f17c32923f00515422b`

Split. `results/split.json` , template hash `cb11391ee245` , seed 0, 14,000 train / 1,000 held-out from databricks-dolly-15k.

Compute. 29 completed training runs: 20 main grid (fast_grid study, 5 methods $\times$ 2 ratios $\times$ 2 seeds) and 9 LR sweep (fast_lr_sweep study, 3 methods $\times$ 3 learning rates $\times$ 1 seed). Total measured wall-clock: **9.1 hours** (4.4 h training + 4.7 h selection), on Kaggle Tesla T4 and NVIDIA RTX 3070.

Exact commands to reproduce from scratch:

```
# 1. Install dependencies
pip install -r requirements.txt

# 2. Verify environment (32 checks, no GPU needed)
python scripts/smoke_test.py

# 3. Frozen split (already committed – only needed once on a fresh clone)
python scripts/make_split.py

# 4. Selection artifacts (CPU; cached after first run)
for m in random perplexity diversity ifd; do
  python scripts/run_selection.py --method $m --ratio 0.05 0.1 --seed 0 1
done
# learning_percentage requires GPU:
python scripts/run_selection.py --method learning_percentage --ratio 0.05 0.1 --seed 0

# 5. Training (GPU required)
python src/runner.py --config configs/fast_grid.yaml
python src/runner.py --config configs/fast_lr_sweep.yaml

# 6. Analysis and figures
python scripts/analyze.py --all
python scripts/figures.py

# All numbers in the paper are produced by steps 6.
# Selection artifacts and run logs are already committed;
# steps 4 and 5 can be skipped to reproduce analysis only.
```